Additive Manufacturing 38 (2021) 101750

Contents lists available at ScienceDirect

Additive Manufacturing

journal homepage: www.elsevier.com/locate/addma

Characterizing material transitions in large-scale Additive Manufacturing☆

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ARTICLE INFO

Keywords:
3D printing
Extrusion
Large-scale
Functionally graded materials
Multi-material

ABSTRACT

Integrating Multiple Materials (MM) into large-scale Additive Manufacturing (AM) is a key for various industrial applications wishing to incorporate site-specific properties into geometrically complex designs that are difficult to manufacture with traditional techniques. Printing with multiple materials is typically accomplished by using layers as natural material boundaries, but having the capability to switch between materials within a single layer without pausing would further expand MM possibilities. This study used Cincinnati Incorporated's Big Area Additive Manufacturing (BAAM) system to explore material transitions with a novel dual-hopper that enables in-situ material blending of a pelletized feedstock. Constructing MM and functionally graded material (FGM) structures requires depositing a specific material composition at a specific geometric location to achieve a desired performance. Accurately implementing this with the BAAM's blended extrusion system requires a thorough understanding of the transition between distinct material compositions. This study characterizes a step-change transition between neat acrylonitrile butadiene styrene (ABS) and carbon fiber-reinforced ABS. Three distinct techniques were compared for analyzing the fiber content, and the transition zone between materials was characterized as a function of transition direction. The transition process was consistent to within 0.7 wt% carbon fiber variation between different layers and prints. The transition between materials was found to be directionally dependent, with ABS to CF/ABS having a transition length of 3.5 m compared to 3.2 m for CF/ABS to ABS. Furthermore, the transition from Material A to Material B was found to be repeatable with a possible variance in transition length of 0.3 m.

1. Introduction

The advantage of Additive Manufacturing (AM) traditionally lies in the ability to construct complex geometries [1,2]. However, most AM systems are restricted to depositing a single material, leading to a performance entirely dependent upon geometry. Recently, newer AM systems have been adapted to deposit multiple materials within a given part, allowing for deliberate placement of each material [3–7]. Through construction of Multi-Material (MM) and Functionally Graded Material (FGM) structures, these advanced systems enable manufacturing of complex geometries with site-specific properties. Typically, MM structures use a Multi-Material Additive Manufacturing (MMAM) system, which utilizes more than one material without pre-mixing, pre-compositing, or non-AM post-processing treatments [8]. FGMs, on the other hand, do not necessarily require an MMAM system. Instead, FGMs have a defined spatial variation in density (single-material) or composition (MM) that achieves a desired functional performance [9]. FGM structures can vary in either a stepwise or continuous manner. A

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https://doi.org/10.1016/j.addma.2020.101750
Received 17 September 2020; Received in revised form 23 November 2020; Accepted 3 December 2020
Available online 5 December 2020
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stepwise FGM consists of multiple regions that differ in composition or microstructure, have discrete boundaries, and are intentionally arranged to achieve a gradient over the entire structure. In contrast, a continuous FGM lacks discrete interfaces and instead has a controlled, smooth gradient in composition or microstructure to attain the desired functionality [10].

MM printing is compatible with a variety of processes, all of which generally assign different materials to different layers for ease of construction. However, bonding dissimilar materials can be difficult, often resulting in structural weak points that are vulnerable to delamination. Traditional sandwich panel designs that are 3D printed typically suffer from fractures at the material interface, regardless of the MMAM method. Vu et al. evaluated sandwich structures with a T-Peel geometry made with a Stratasys PolyJet Connex 350 and observed fractures favored the material interfaces [3]. While aligning the interfaces perpendicular to the build direction increased their fracture resistance, interface failure was still common [11]. Brischetto et al. experienced similar behavior on a multi-nozzle Fused Filament Fabrication (FFF) system. Exchanging acrylonitrile butadiene styrene (ABS) for polylactic acid (PLA) in the face sheets yielded a weaker sandwich structure despite maintaining the same PLA honeycomb core. Once again, delamination between the dissimilar materials was largely responsible [4]. Roger and Krawczak demonstrated that simply adding a filler can influence bonding and delamination tendencies. An interpenetrating arrangement preventing the material interface from occupying a single plane improved mechanical performance, but the single-material counterpart still showed a greater resistance to delamination [12]. Kim et al. also investigated interface location and number by placing multiple interfaces in both the print and z-directions (across layers). While z-direction boundaries decreased void formations, boundaries in the build-direction resulted in voids that delaminated during testing [13].

MMAM processes have routinely utilized layers as inherent discrete boundaries when implementing stepwise FGMs. In a continuation of previous studies, Vu et al. assessed the effect of a stepwise FGM core on a sandwich structure performance. Resistance to delamination in T-Peel testing demonstrated an increased resistance to peel and fracture compared to the typical single-material cores [11]. To balance structural flexibility and rigidity, researchers implemented a similar stepwise FGM approach for construction of a robot shell. Numerous material changes gradually increased the elastic modulus from 1 MPa to 1 GPa, allowing it to survive impacts that shattered a rigid version while limiting the elastic deformation that made the flexible version inoperable [5]. While boundary-representation approaches struggled to place MM within the same layer, a new FGM design enabled this function via volumetric placement. It successfully demonstrated control of porous regions and differing materials [6]. One in-depth study used voxel-based inkjet printing techniques to demonstrate the strength gains provided by continuous FGMs structures compared to stepwise variants [14], while another utilized sequential MMAM to construct FGMs ranging in size from millimeters to meters [15]. Other bioinspired FGMs have targeted medical usage in orthopedic implants [16,17]. Regardless of the process or materials, there are a wide variety of applications and interests in using MMAM to create FGM structures with unique properties [18–20]. In each case, FGMs functioned as a bridge between dissimilar materials, enabling regional properties while limiting typical MM defects such as delamination. As such, it is critical to understand the options available to AM for FGMs and to explore the best production methods at all available scales. The goals of this study were to conduct the first investigation of in-situ blending of discrete materials using LS-ME, to demonstrate continuous transition from one material to another during printing, and to discuss the application of this technique to FGM printing.

2. System description

The development of MM and FGM technologies has grown to include large-format AM technologies such as Cincinnati Inc.'s Big Area Additive Manufacturing (BAAM) system. Early efforts to take advantage of its pelletized feedstock and extrusion screw to incorporate in-situ material switching [21–24] has culminated in the development of a dual-hopper feeding system with automated material control [7,25]. Shown in Fig. 1, the system provides a unique ability to change materials on-the-fly and enables printing of MM and FGM structures without the need for discrete material boundaries or print stoppage.

The inherent mixing within a single-screw extruder causes a step-change in material feedstock to deposit as a blended region, or transition zone, of continuous composition change between two materials A and B. When tracked as a function of distance or time, the composition of the deposited material will follow a distinctive "transition curve" from A to B. Fig. 2 illustrates some potential transition curves if the step-change in feedstock were executed at time t_0. The idealized step-change, in red, reflects absolutely no material mixing prior to extrusion and occurs at time t_r, which represents the time required for deposition of the residual volume of Material A, V_{RA}, remaining in the barrel after t_0. Graphically, V_{RA} can be represented as the area under the red line and to the left of time t_r. A simple linear model, in green, represents a single-rate shift from Material to B. Conservation of mass requires that the same residual volume of Material A exits the extruder, so the area under the green line must still equal V_{RA}, which also indicates an equal composition at t_r. While the slope of the line may vary, there will always be an initiation time, t_i, and time of steady-state Material B, t_s. Various conditions could also produce non-linear transitions between materials. For instance, a screw geometry intended to avoid mixing could see limited interaction at the material interface and shorten the transition zone, as conveyed by the blue line, while a reduction in viscosity may cause an early transition, as shown by the gray line. Even though t_s would change, the area under the curve would still equal that of the shaded red area, indicating t_s cannot be less than t_r.

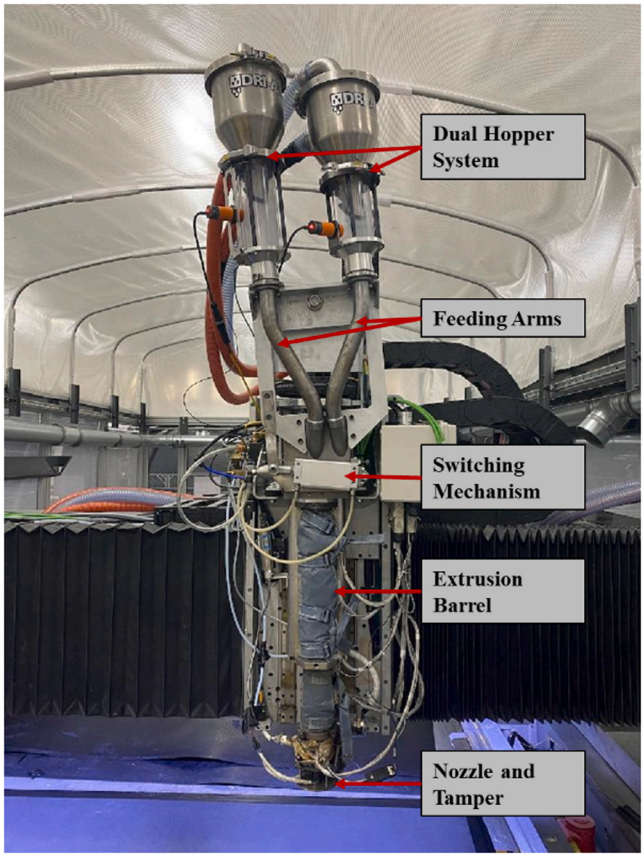


Fig. 1. Dual-hopper mechanism installed on the BAAM system.

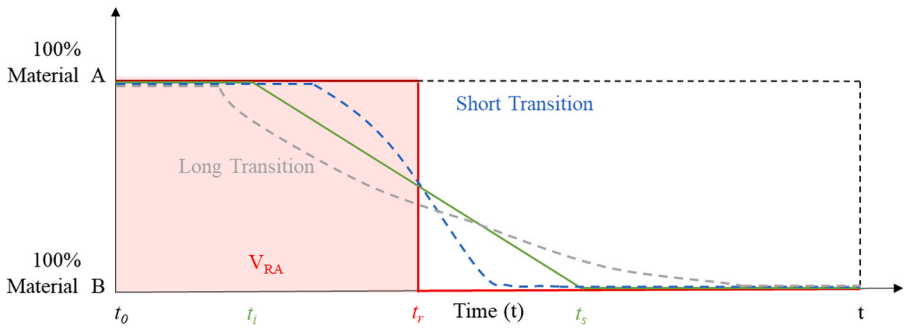


Fig. 2. A schematic depicting the change in composition over an arbitrary Time, t. The shaded red square, V_{RA}, represents the volume of residual Material A remaining in the barrel of the extruder. The solid lines represent theoretical Step (Red) and Linear (Green) functions. The two dotted lines represent the possible non-linear transitions (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

While time is an important consideration for AM processing, the transition should be considered as a function of deposited material to best asses how it can be included in and will impact a printed part. Fig. 3 illustrates how the physical transition length could differ with direction and influence part deposition strategies. Fig. 3a shows the transition from Material A (gray) to Material B (white) on layers 1 and 3 then reverses the direction layers 2 and 4. In this situation, the transition behavior is direction dependent and leads to variation in L_i (distance traveled where composition first changes) and L_s (distance at which transition is complete) in alternating layers. Even with the transition centered and L_0 (point of feedstock switch) occurring the same distance from layer ends P_A and P_B, the difference in transition length leads to significantly different compositional gradients. Each direction is given its own variable, i.e. $L_{0\to B}$ is the point of feedstock switch when transitioning to Material B. Fig. 3b plots material composition as a function of total distance traveled along two successive beads. While $L_{0\to A}$ and $L_{0\to B}$ are equidistant from P_A and P_B, respectively, the distance from $L_{i\to A}$ to $L_{s\to A}$ is shorter than from $L_{i\to B}$ to $L_{s\to B}$. This is due both to $L_{i\to B}$ beginning earlier than $L_{i\to A}$ and to $L_{s\to B}$ taking longer to reach than $L_{s\to A}$. The resulting nonuniform structure has that can lead to unanticipated distribution of stress and reduced mechanical performance of the final part.

Several printing parameters impact print quality when using BAAM, and many are material specific. For instance, BAAM uses a five-zone temperature profile to achieve a steady melt that is tailored to each material to ensure high-quality deposition. Since the dual-hopper introduces a bi-material system, it will be important to choose thermally, and otherwise, compatible materials. Duty et al. determined that material "printability" relies on achieving a certain level of viscous flow without exceed pressure limits, indicating that typical screw speed should be considered along with melt temperature when assessing material compatibility [26]. Similarly, the BAAM utilizes a range of screw geometries and screw speeds that could influence the mixing behavior within the barrel, altering the transition zone, as initially seen in Sudbury et al. [23]. Material transitions will also be affected by thermo-physical and rheological properties through their impact on mixing behavior. For example, materials reinforced with fillers like glass or carbon fibers have a significantly higher viscosity and storage modulus than the unreinforced matrix materials [27–29]. Given the variety of factors at play, compromises between the typical print parameters of two materials may be necessary to maintain printable conditions for both materials and avoid freezing or bonding issues during deposition [26].

3. Experimental

Material compositional analysis was then used to assess the reliability of the dual-hopper printing process by comparing fiber content at

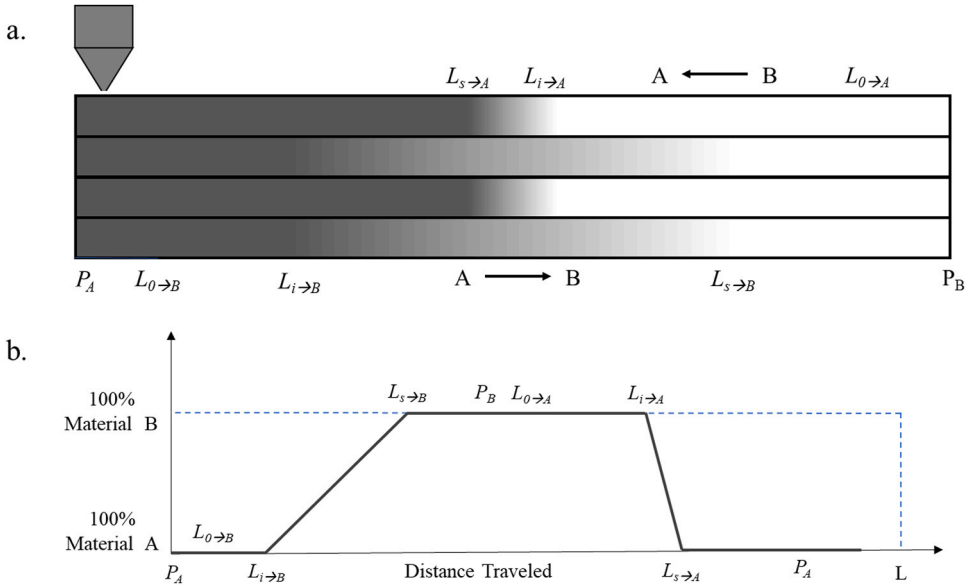


Fig. 3. a. A schematic of a four-layer bead with transitions limited to one direction per layer where gray is Material A and white Material B. b. The change in material composition as a function of distance traveled for the first two layers.

similar locations across different layers and printed parts. Compositional analysis also functioned as a tool for constructing transition curves that described the progression from Material A to Material B over distance and time. Two additional characterization techniques were evaluated, and their reliability was compared to the original technique. The advantages and limitations of each were discussed in the context of future investigations.

3.1. Constituent content analysis

The transition between materials was tracked using material compositional analysis. In this instance, the primary difference in material composition was the carbon fiber content, so the weight percentage of carbon fiber in each sample was measured to determine the relative material composition. While the CF/ABS was listed as having 20 wt% fiber content, the actual weight may vary within a given batch. As such, the raw feedstock was analyzed to establish a baseline for the individual batch that was used in these experiments. Each specimen's fiber content was then compared against this batch-specific baseline to determine the relative amount of CF/ABS present. A similar baseline was defined for neat ABS by measuring the amount of residue that contains no carbon fiber. A Mettler Toledo Balance, XS204 Deltarange with 0.1 mg accuracy was used for all mass measurements. The three techniques below were used to characterize the dual-hopper system's repeatability and consistency.

3.1.1. Carbonization in nitrogen (CIN)

Following ASTM D3171-15 Procedure H: "Matrix Carbonization in a Nitrogen-Purging Furnace", this process heated the samples in an inert nitrogen atmosphere to thermally degrade the polymer matrix [30]. A similar investigation demonstrated the effectiveness of the nitrogen atmosphere with fibers retaining 99 wt% of original mass after 60 min at 600 °C [31]. In some cases, a residual amount of non-fiber mass remained after the procedure, so a neat ABS feedstock sample was used as a control in each sample set to define a baseline for the mass of the fiber-free residue. This control was used to calculate a ratio of residue mass to original mass that was then used to appropriately adjust the calculated fiber content for each sample present in that experimental set, which is detailed in [25]. CIN used a Lindberg Blue M nitrogen tube furnace held at 565 °C for five hours while flowing nitrogen. CIN was chosen as the primary characterization method used in this study due to its excellent reliability [31].

3.1.2. Carbonization in air (CIA)

Following ASTM D3171-15 Procedure G: "Matrix Burn-off in a Muffle Furnace", this process eliminated the polymer matrix via combustion in an ambient atmosphere, leaving behind a mix of ash and fiber reinforcement [30]. The variety of carbon fibers available leads to a disparity in what processing temperature levels may damage or degrade the fibers, but a general limit of 500–550 °C proved acceptable for most instances [32]. Similar to CIN, CIA utilized the ratio of the residue from a control sample to the residue of a given sample to calculate the portion of the residue that was carbon fiber. CIA used an MTI KSL-1100X-S-UL Compact Muffler Furnace at 450 °C for four hours. A dwell time of four hours resulted in less than 1% of residual weight from a neat ABS sample, with negligible reduction over longer durations.

3.1.3. Ultrasonic-assisted acid digestion (UAAD)

Based on ASTM D3171-15, this procedure selectively digested the polymer matrix with an acidic solution. The standard ASTM procedure uses hydrogen peroxide as an accelerant, but the current study used sonication to mechanically agitate the samples. Test tubes containing HPLC Acetone ($\geq$ 99.5% purity) and a sample remained in the CPX3800 Ultrasonic Bath until 20 min after no visible traces of the polymer matrix remained (typically 2 h). The solution was passed through Whatman Grade GF/A glass microfiber filters (47 mm diameter with 1.6-micron pores) and rinsed with ultra-pure milli-q water to separate the fibers from liquid solution. After drying overnight at 80 °C, the samples were weighed to determine the residual fiber mass.

CIA and UAAD were used to characterize consecutive samples form printed specimens to analyze their accuracy, and both techniques were then compared to CIN to identify the best process for future investigations based on compatibility, versatility, accuracy, ease of use, and time investment. Compatibility will be largely dependent on the characteristics of the two materials involved. Differences in degradation temperatures or temperature resistant fillers would enable CIN and CIA while a difference in chemical vulnerability or chemically resistant fillers would allow for UAAD. If more than one technique was relevant, their demonstrated accuracy would function as the next deciding factor. Given a comparable accuracy, ease of use and time investment will play the final role in technique choice. For example, UAAD equipment is readily available and, in general, could test more samples per hour than either CIA or CIN, though it required more manual interaction.

3.2. Sample preparation

All samples were printed using the dual-hopper system installed on BAAM – B at ORNL's Manufacturing Demonstration Facility. Techmer Engineered Solutions supplied two grades of ABS in pelletized form: a neat ABS (HIFILL ABS 1000 3DP) and a 20 wt% CF/ABS (Electrafil J-1200/CF/20/ 3DP). Feedstock was dried at 85 °C prior to printing.

To ensure that transitions between Material A (CF/ABS) and Material B (ABS) could be completed in a single layer, a single-walled rectangle (Fig. 4) was printed with dimensions 3.50 m by 1.52 m (11.5 ft by 5 ft). Guided by initial trials, the step changes in pelletized feedstock material (i.e. L_0/t_0) were placed at opposite corners (1 and 3) of the rectangle so that the residual volume, V_{RA}, was primarily extruded on the short sides of the rectangle and the transition between materials was roughly centered along the long sides. Each sample set consisted of a three-step printing process with no purging of material between steps. Initially, a single layer of the rectangle was deposited for baseline analysis. Then a 20-layer rectangle (layer height of 6.35 mm and single-bead width of 15.24 mm, resulting in dimensions of 127 mm by 15.24 mm) was deposited to characterize the consistency of transitions across multiple layers. Finally, a single layer was printed that was similar to the first to assess any compounded effects of multiple transitions within a print. This three-step printing process was conducted three times to analyze the repeatability of the hopper switch across identical structures and conditions. The material transitions were characterized using compositional analysis and the consistency of the transition region from layer-to-

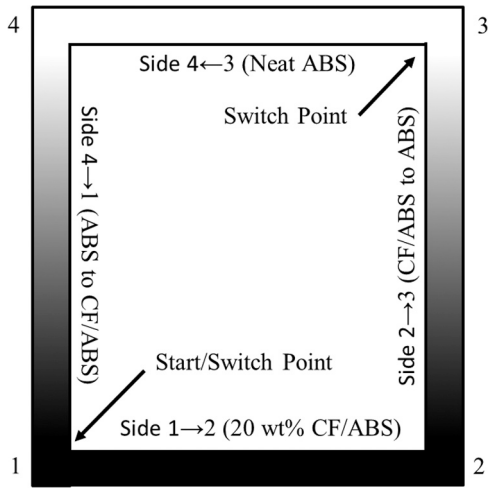


Fig. 4. Rectangular geometry printed on BAAM – B showing transition points and blended regions.

layer and print-to-print was examined. Additionally, both transition directions, A → B and B → A, were compared to observe any differences caused by transition direction and to determine the reversibility of the process.

Each print used a nozzle diameter of 10.16 mm (0.4 in), layer height of 6.35 mm (0.25 in), bead width of 15.24 mm (0.6 in), and the same screw with non-mixing geometry operating at 300 RPM. The resulting travel speed was 106.7 mm/s (4.2 in/s). Thermal conditions maintained a melt temperature of 245 °C and bed temperature of 100 °C, typical deposition temperatures for both materials.

The procedure for obtaining samples from the printed rectangles was consistent across all measurement techniques. Each single-layer rectangle was sectioned into 152.4 mm (6 in) rods with labels indicating the order and print direction. Individual, 8.4 mm long specimens were cut from the rods using an Isomet 1000 Precision Saw. The relative position of the samples was defined as the distance traveled from the last hopper switch location. Eight locations were initially selected to from around the rectangle to capture both single-material and transition regions when transitioning from Material A to Material B and back to Material A. Fig. 5 shows the relative location of samples for a single-layer rectangle. It should be noted that color was a poor indicator of the relative carbon fiber content, as even the samples that were predominantly neat ABS (upper right side of Fig. 5) were slightly black in appearance.

3.3. System repeatability and transition curves

Using CIN, system repeatability was evaluated by measuring the consistency of the sampling technique for adjacent samples, transition repeatability between layers, and transition repeatability among prints. Testing of three consecutive (essentially identical) samples at each location in Fig. 5 enabled two different observations: the samples in the single-material regions allowed for an analysis of technique precision while those in the transition region would then serve as an assessment of compositional variance over a small volume. The effects of numerous hopper switches within a single print on transition behavior was assessed by comparing average fiber content of the first and final layers. Comparisons of three separate but identical prints were made to evaluate potential inconsistencies in overall transition behavior caused by the dual-hopper switch.

After assessing the consistency of the process, CIN was also used to construct a transition curve for both directions: CF/ABS to ABS and ABS to CF/ABS. Selective measurements throughout the length of the printed transition were plotted as a function of both distance and time to demonstrate the change in composition as the print progressed. The two resulting transitions were compared to understand how the direction of the material change may impact the transition behavior. Furthermore, CIA and UAAD were each used to construct a transition curve from samples immediately adjacent to those analyzed in the CIN CF/ABS to ABS transition curve.

4. Results and discussion

4.1. Baseline fiber content

Analysis of the as-received CF/ABS pellets using CIN determined a baseline average fiber content among five sample sets of 22.0 wt% with a standard deviation of 0.2 wt%. Literature showed that CIN typically had a deviation of ≤ 1 wt% from the composite's stated fiber content [31]. Given the experimental repeatability and literature support, the measured 22.0 wt% carbon fiber content was treated as a baseline for the provided CF/ABS feedstock instead of the advertised 20 wt%.

4.2. Consistency of CIN

Fig. 6 depicts the fiber content as measured by CIN of each of the eight locations shown in Fig. 5. Three consecutive samples from each spot were measured to obtain averages and standard deviations. The residual mass for neat samples varied such that calculated fiber content could be negative due to slight degradation differences between specimens and the control. Thus, any negative results were treated as having a 0 wt% fiber loading. The only two such instances had a standard deviation of less than 0.5 wt%. As shown in Fig. 6a, each set of samples in the single-material region exhibits a low standard deviation, which indicated an accurate testing method. Similarly, the transition specimens in Fig. 6b exhibited standard deviations low enough to consider the compositional gradient consistent. As such, a single 8.4 mm sample represented, at least, a 25.4 mm surrounding area. The sample locations in Fig. 6b were chosen to illustrate that the geometric middle, Position Two, is a poor indicator of the transition zone, and so Position One was visually selected to ensure placement within the transition region and was not the same for CF/ABS to ABS as ABS to CF/ABS.

4.3. Consistency among layers within a print

CIN measurements at layers one and twenty of each print were averaged for each location and charted in Fig. 7. Transition samples were at different locations (1.98 m vs 2.59 m) for each direction due to differences in the transition behavior that rendered the midpoints of the transitions not equal and necessitated visual selection mentioned for Fig. 6b. Fig. 7a and b shows consistent measurements with minimal deviation outside of the transition regions. Unlike Fig. 7b, 7a had a significant discrepancy in fiber content in the transition region. Lower fiber loading at later layers in that transition indicated a quicker transition and, thus, an overall shift in the initially defined mixed

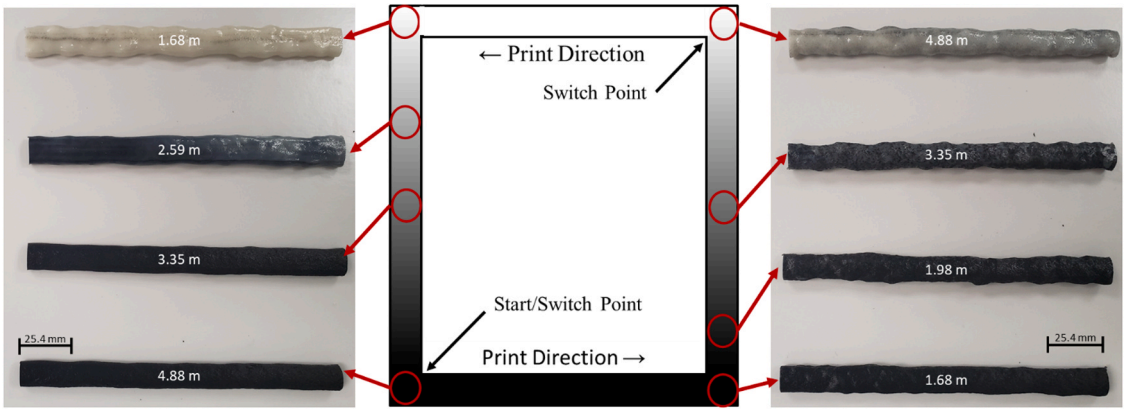


Fig. 5. Depiction of sampling locations for the process repeatability portion of the study. The number in the center of each rod measures the distance from hopper switch.

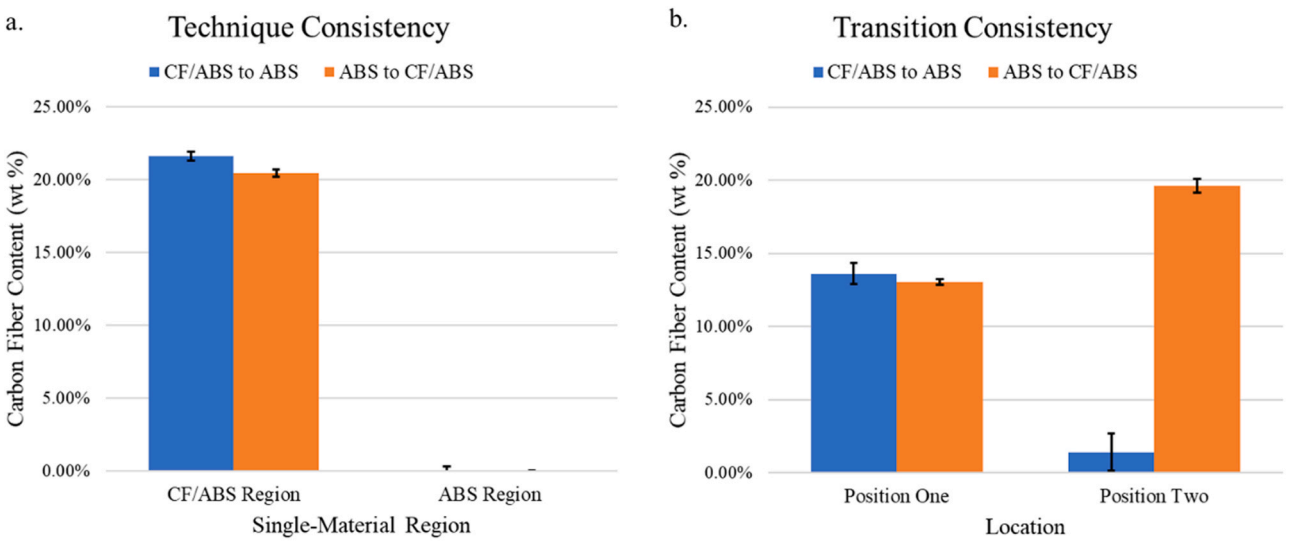


Fig. 6. Average carbon fiber content of three consecutive samples for four locations from each transition direction. a. The single-material regions. b. The transition region.

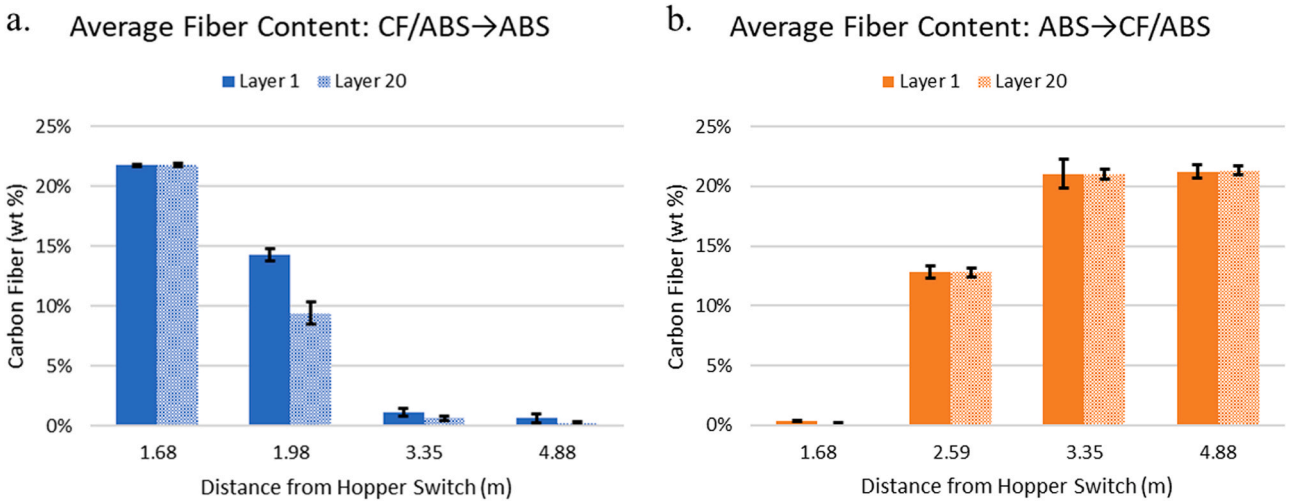


Fig. 7. Measurements were taken for each of three prints and averaged for each location. The average fiber content for layers one and 20 are compared for both a. the CF/ABS to ABS transition and b. the ABS to CF/ABS transition.

composition region. As such, the transition direction appears to have a significant impact on the transition region after multiple hopper switches were made during a single print with fiber reinforcements in one material. This could be explained by the build-up of residual fibers within the barrel of the extruder that aid in expediting the transition. Similarly, later results showed that even after fiber content drops below 1.5% and reaches steady-state behavior, the neat ABS regions still had a visually observable fiber presence. While this could be caused by residual fibers, the exact mechanism responsible for this behavior was not clear.

4.4. Consistency among multiple prints

Fig. 8 shows average fiber content for CIN measurements of three separate prints, using one sample from each. There was minimal variation at each location, regardless of transition direction. All standard deviations were less than 0.7 wt%. For the transition zone, the standard deviation was 0.59 wt% in the ABS to CF/ABS direction and 0.63 wt% in the CF/ABS to ABS direction. The largest observed standard deviation was 0.6 wt% at 3.35 m from hopper switch in the ABS to CF/ABS transition. As such, the transition appeared to be repeatable across multiple prints of the same structure, suggesting that transition behavior can be characterized using a single print.

4.5. Transition curves

Since the in-situ material switching capability was reproducible, characterization of a single transition region for both directions attained reliable transition curves. Content analysis using CIN enabled the creation of a complete transition curve in both transition directions, including data points from before and after the transition zone. Once again, distance and time from the hopper switch were used to track the transition. The CF/ABS to ABS transition shown in Fig. 9 was marked with three distinct zones. In the "Purge Zone," there was no noticeable change after the hopper switch, which resulted from material already present in the barrel that must cycle through before the introduced material appeared. The "Transition Zone" began at L_i where the next data point shifted by at least 1 wt% carbon fiber. Generally, this phenomenon was easily identified due to a sharp increase or decrease of the fiber content. The Transition Zone exhibited a constant change in

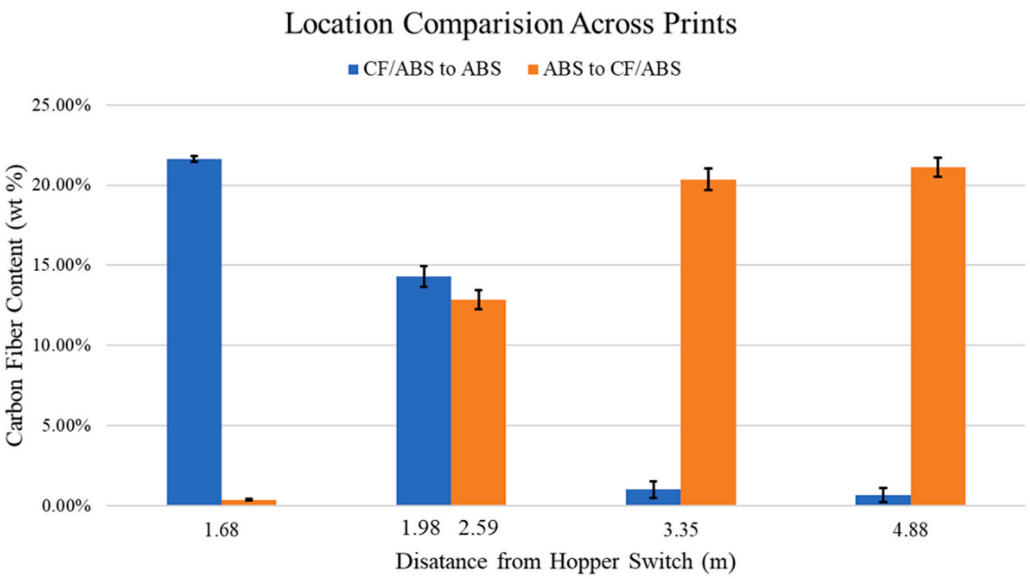


Fig. 8. The carbon fiber content at each CIN testing location for both transition directions. Note, the second testing location had different distances, as described above.

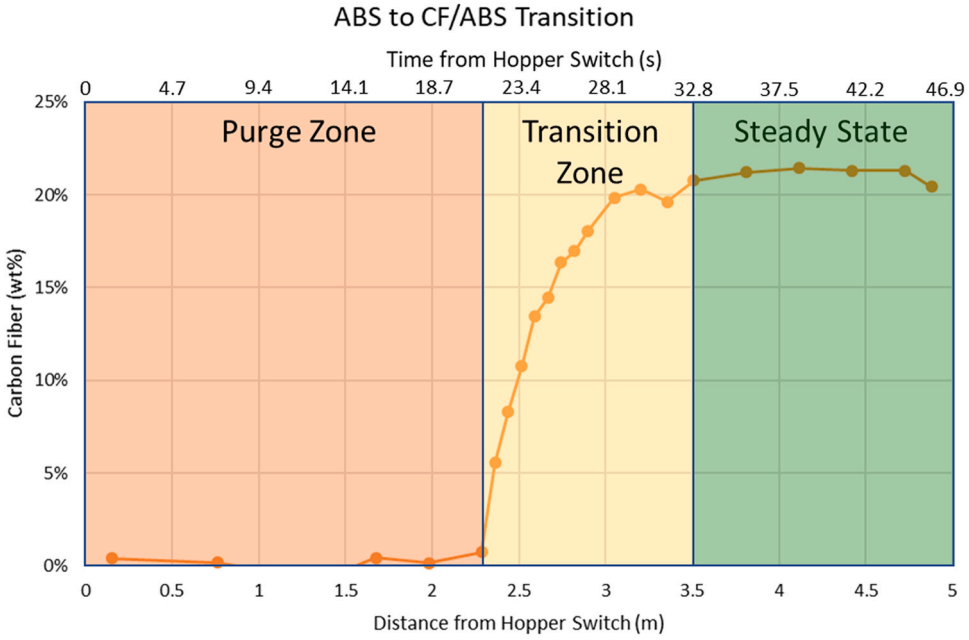


Fig. 9. Fiber content vs both time (s) and distance (m) across the neat ABS to CF/ABS transition. Sampling rate was increased in the Transition Zone to improve resolution as fiber content rapidly increased.

composition characteristic of continuous FGMs. Two different criteria quantified the beginning of the "Steady-State Zone," or L_s. Steady-State for CF/ABS began at the point where both it and the following point return a carbon fiber content above 20 wt%, as seen in Fig. 9. Steady-State for neat ABS was achieved when carbon fiber content falls below 1.5 wt%.

Fig. 10 places both curves on a single graph without the zones for visual clarity. It highlights the differences in behavior, clearly showing that the onset of the Transition Zone was sooner for CF/ABS to ABS. With different starts for their Transition Zones, the CF/ABS and ABS material pair did not qualify as truly reversible, which indicated that the direction of the material transition should be considered as a parameter of interest when evaluating transition behavior. Table 1 displays data for both transitions from Fig. 10. Since print speed varied as a function of chosen screw speed, establishing both distance and time from hopper switch allows for direct comparison to planned future experiments. Maintaining a consistent bead shape required that travel speed increase with an increase in flow rate created by a higher screw speed. As such, time does not provide a direct comparison. The "Combined" row summed the Purge and Transition Zones to emphasize the total time and distance necessary to reach steady-state for the new material after a hopper switch. Looking at Combined times, CF/ABS to ABS transitions 2.9 s quicker. However, it derived the advantage solely from a 4.3 s faster Purge Zone, as ABS to CF/ABS contained a 1.5 s faster Transition Zone. Since the Purge Zone's length determined where a material transition began relative to initiation of the hopper switch, accurate design and implementation of MM and FGM printing necessitates an understanding of its characteristics.

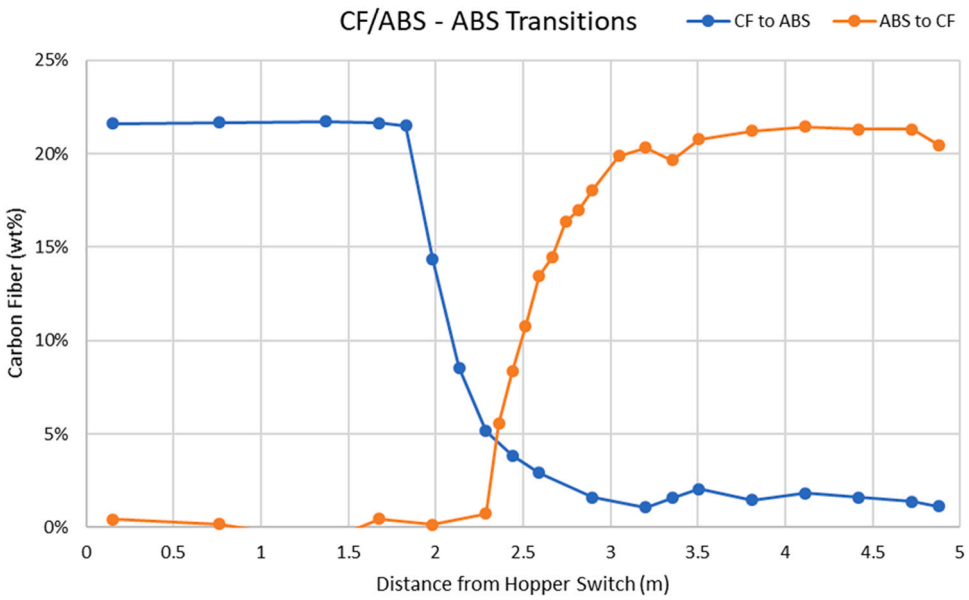


Fig. 10. A combination of both transition curves that more easily demonstrates their differences in transition behavior. The singular dip below 0 wt% was less than 1 wt%, so it was attributed to the error inherent to the CIN process, as shown in [31].

Table 1
Numerical comparison of Purge and Transition Zones.

Direction	CF/ABS ABS		ABS CF/ABS	
Unit	Meters	Seconds	Meters	Seconds
Purge Zone	1.83	17.1	2.29	21.4
Transition Zone	1.37	12.9	1.22	11.4
Combined	3.20	30.0	3.51	32.9

4.6. Comparison of constituent content analysis methods

While CIN had been previously validated as a reliable technique [30,31], assessing additional characterization options provided the opportunity to discuss which would be best suited for future work. CIA and UAAD were used to construct shape curves using samples adjacent to those used in CIN. Fig. 11 contains transition curves constructed using all three fiber content measurement techniques. All three methods returned similar fiber content measurements for each location. At every location, CIA measurements were, on average, 1.1 wt% lower than CIN measurements, most likely due to oxygen's presence allowing for complete combustion of the ABS matrix as opposed to a nitrogen atmosphere ensuring some amount of residue is present. This effect escalated in neat ABS regions, where combustion led to less residue compared to CIN and miniscule variation moved printed samples below the control. UAAD consistently returned a value between the two carbonization techniques, which indicated it may be the most reliable method for the CF/ABS and ABS material pair.

5. Conclusions and future work

BAAM-B's dual-hopper system and in-situ material switching capability have the potential to create FGM and MM structures using LS-ME. Constituent content analysis provides a valuable tool for tracking the transition between materials with different reinforcements, which allows for in-depth studies on the influential parameters and toolpath

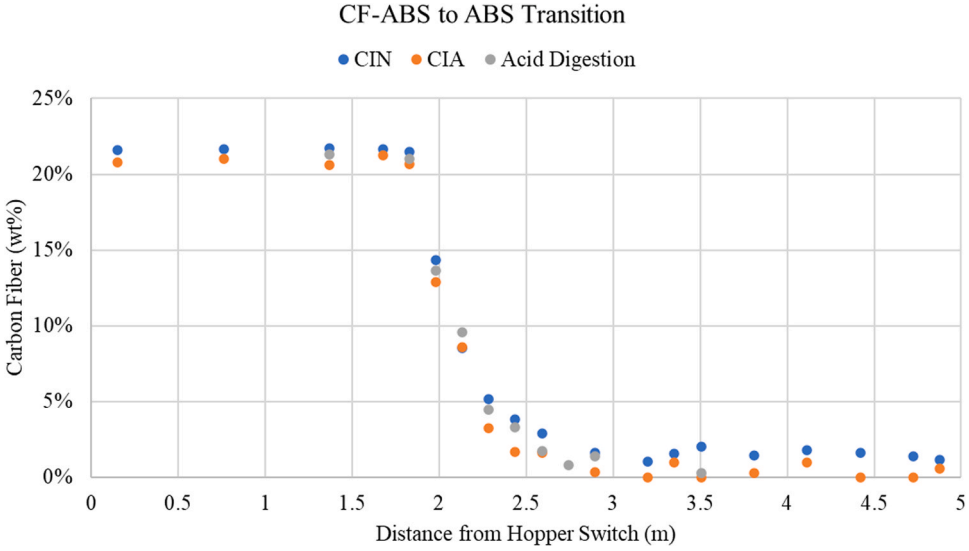


Fig. 11. Characterization of the fiber content during the CF/ABS to ABS transition using CIN, CIA, and the developed UAAD methods.

planning to achieve site-specific properties. This study documented the positional composition for step-change transitions between ABS and CF/ABS in both directions using a variety of techniques. All three techniques produced comparable fiber content measurements and the dual-hopper system was shown to produce a consistent material composition across multiple prints during the initial layers. Three distinct zones were evident in each transition curve: a Purge Zone, a Transition Zone, and Steady State. With the given print parameters, the Purge Zone was about 2 m and the Transition Zone was roughly 1.25 m for the materials used. However, the length of these zones varied within ~ 10% based on the direction of the material transition. The Purge Zone for CF/ABS → ABS was shorter than ABS → CF/ABS, but the Transition Zone was slightly longer. Detailed characterization of these transition regions is critical for proper toolpath planning of multi-material systems.

In future studies, different material pairs will be investigated to determine the effects of varying rheological properties and reinforcement content. Processing parameters, such as screw design and speed, will also be evaluated to determine their influence among the different material pairs. The influence of material properties and processing parameters on the dual-hopper transition curves will inform the design and process planning of FGM components made using LS-ME.

CRediT authorship contribution statement

James Brackett: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization, Supervision, Project administration. **Yongzhe Yan:** Methodology, Investigation, Data curation, Resources. **Dakota Cauthen:** Methodology, Validation, Investigation, Data curation. **Vidya Kishore:** Conceptualization, Writing - review & editing. **John Lindahl:** Investigation, Resources. **Tyler Smith:** Investigation, Resources. **Zeke Sudbury:** Methodology, Investigation. **Haibin Ning:** Methodology, Resources, Supervision. **Vlastimil Kunc:** Conceptualization, Resources, Supervision, Funding acquisition. **Chad Duty:** Conceptualization, Methodology, Resources, Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

One of the authors of this article is part of the Editorial Board of the journal. To avoid potential conflict of interest, the responsibility for the editorial and peer-review process of this article lies with a different Editor, Eric MacDonald. Furthermore, the authors of this article were removed from the peer review process and had no access to confidential information related to the editorial process of this article.

Acknowledgements

Research sponsored by the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Industrial Technologies Program, under contract DE-AC05-00OR22725 with UT-Battelle, LLC. This research was supported by the DOE Office of Energy Efficiency and Renewable Energy, Manufacturing Demonstration Facility. The authors are grateful for the equipment and assistance provided by Cincinnati Incorporated and for the materials supplied by Techmer Engineered Solutions. The University of Alabama Birmingham's Materials Science Department provided assistance with testing and data collection. The authors are also would like to thank the Oak Ridge Institute for Science and Education for its contributions through the Higher Education Research Experiences Program.

References

[1] J. Jiang, X. Xu, J. Stringer, Support structures for Additive Manufacturing: a review, J. Manuf. Mater. Process. 2 (4) (2018) 64. ⟨https://www.mdpi.com/2504-4494/2/4/64⟩.

[2] J. Jiang, A novel fabrication strategy for Additive Manufacturing processes, J. Clean. Prod. 272 (2020) 122916, https://doi.org/10.1016/j.jclepro.2020.122916.

[3] I. Vu, L. Bass, N. Meisel, B. Orler, C. B. Williams, D.A. Dillard, Characterization of mutli-material interfaces in PolyJet Additive Manufacturing, in: Proceedings of the Solid Freeform Fabrication Symposium Proceedings, 2015, pp. 959–982.

[4] S. Brischetto, C. Ferro, R. Torre, P. Maggiore, 3D FDM production and mechanical behavior of polymeric sandwich specimens embedding classical and honeycomb cores, Curved Layer. Struct. 5 (2018) 80–94, https://doi.org/10.1515/cls-2018-0007.

[5] N.W. Bartlett, M.T. Tolley, J.T.B. Overvelde, J.C. Weaver, B. Overvelde, K. Bertoldi, G.M. Whitesides, R.J. Wood, A 3D-printed, functionally graded soft robot powered by combustion, Science 349 (6244) (2015) 161–165, https://doi.org/10.1126/science.aab0129 (in English).

[6] B. Ezair, G. Elber, Fabricating functionally graded material objects using trimmed trivariate volumetric representations, in: Proceedings of the Fabrication and Sculpting Event, 2017.

[7] T. Smith, A.A. Hassen, R. Lind, J. Lindahl, P. Chesser, A. Roschli, V. Kumar, V. Kishore, B. Post, J. Failla, C. Duty, L. Love, V. Kunc, Dual material system for polymer large scale Additive Manufacturing, in: Proceedings of the Society for the Advancement of Material and Process Engineering 2020, Seattle, WA, May 4–7, 2020.

[8] M. Vaezi, S. Chianrabutra, B. Mellor, S. Yang, Multiple material additive manufacturing – part 1: a review, Virtual Phys. Prototyp. 8 (1) (2013) 19–50, https://doi.org/10.1080/17452759.2013.778175.

[9] G.H. Loh, E. Pei, D. Harrison, M.D. Monzón, An overview of functionally graded additive manufacturing, Addit. Manuf. 23 (2018) 34–44, https://doi.org/10.1016/j.addma.2018.06.023.

[10] G. Udupa, S.S. Rao, K.V. Gangadharan, Functionally graded composite materials: an overview, in: S. Narendranath, M.R. Ramesh, D. Chakradhar, M. Doddamani, S. Bontha (Eds.), Proceedings of the International Conference on Advances in Manufacturing and Materials Engineering, vol. 5, (Procedia Materials Science). Elsevier Science Bv, Amsterdam, 2014, pp. 1291–1299.

[11] I.Q. Vu, L.B. Bass, C.B. Williams, D.A. Dillard, Characterizing the effect of print orientation on interface integrity of multi-material jetting additive manufacturing, Addit. Manuf. 22 (2018) 447–461, https://doi.org/10.1016/j.addma.2018.05.036.

[12] F. Roger, P. Krawczak, 3D-printing of Thermoplastic Structures by FDM using Heterogeneous Infill and Multi-materials: An Integrated Design-advanced Manufacturing Approach for Factories of the Future, 2015.

[13] H. Kim, E. Park, S. Kim, B. Park, N. Kim, S. Lee, Experimental study on mechanical properties of single- and dual-material 3D Printed Products, Proc. Manuf. 10 (2017) 887–897, https://doi.org/10.1016/j.promfg.2017.07.076.

[14] M.J. Mirzaali, A. Herranz de la Nava, D. Gunashekar, M. Nouri-Goushki, R.P. E. Veeger, Q. Grossman, L. Angeloni, M.K. Ghatkesar, L.E. Fratila-Apachitei, D. Ruffoni, E.L. Doubrovski, A.A. Zadpoor, Mechanics of bioinspired functionally graded soft-hard composites made by multi-material 3D printing, Compos. Struct. 237 (2020) 111867, https://doi.org/10.1016/j.compstruct.2020.111867.

[15] N.A. Lee, R.E. Weber, J.H. Kennedy, J.J. Van Zak, M. Smith, J. Duro-Royo, N. Oxman, Sequential multimaterial additive manufacturing of functionally graded biopolymer composites, 3D Print. Addit. Manuf. 7 (5) (2020) 205–215, https://doi.org/10.1089/3dp.2020.0171 (in English).

[16] N. Kladovasilakis, K. Tsongas, D. Tzetzis, Finite element analysis of orthopedic hip implant with functionally graded bioinspired lattice structures, Biomimetics 5 (3) (2020) 44, https://doi.org/10.3390/biomimetics5030044 (in English) (Art no. 44).

[17] P. Heinl, L. Muller, C. Korner, R.F. Singer, F.A. Muller, Cellular Ti-6Al-4V structures with interconnected macro porosity for bone implants fabricated by selective electron beam melting, Acta Biomater. 4 (5) (2008) 1536–1544, https://doi.org/10.1016/j.actbio.2008.03.013 (in English).

[18] M. Naebe, K. Shirvanimoghaddam, Functionally graded materials: a review of fabrication and properties, Appl. Mater. Today 5 (2016) 223–245, https://doi.org/10.1016/j.apmt.2016.10.001 (in English).

[19] R.M. Mahamood, E.T. Akinlabi, Types of Functionally Graded Materials and Their Areas of Application. Functionally Graded Materials, Springer International Publishing, Cham, 2017, pp. 9–21.

[20] A. Bandyopadhyay, B. Heer, Additive manufacturing of multi-material structures, Mater. Sci. Eng.: R: Rep. 129 (2018) 1–16, https://doi.org/10.1016/j.mser.2018.04.001.

[21] Z. Sudbury, C. Duty, K. Vlastimil, Expanding material property space maps with functionally graded materials for large scale Additive Manufacturing, in: Proceedings of the Solid Freeform Fabrication Symposium Proceedings, 2017, pp. 459–484.

[22] Z. Sudbury, C. Duty, V. Kunc, V. Kishore, C. Ajinjeru, J. Failla, J. Lindahl, Characterizing material transition for functionally graded material using Big Area Additive Manufacturing, in: Proceedings of the Solid Freeform Fabrication Symposium, August 8, 2016, Austin, TX, 2016, pp. 738–747. ⟨https://sffsymposium.engr.utexas.edu/sites/default/files/2016/058-Sudbury.pdf⟩.

[23] Z. Sudbury, C. Ajinjeru, V. Kishore, C. Duty, P. Liu, V. Kunc, Blending of fiber reinforced materials using Big Area Additive Manufacturing, in: Proceedings of the Society for the Advancement of Material and Process Engineering, May 22–25 2017, Seattle, WA, 2017.

[24] J. Brackett, Y. Yan, D. Cauthen, V. Kishore, J. Lindahl, T. Smith, H. Ning, V. Kunc, C. Duty, Development of functionally graded material capabilities in large-scale extrusion deposition Additive Manufacturing, in: Proceedings of the Solid Freeform Fabrication Symposium, Austin, TX, 2019, pp. 1793–1803.

[25] J. Brackett, D. Cauthen, T. Smith, V. Kunc, E. Duty Chad, The influence of processing parameters on the transition zone for blended material 3D printing, in: Proceedings of the Society for the Advancement of Material and Process Engineering, May 4–7 2020, Seattle, WA, 2020.
[26] C. Duty, C. Ajinjeru, V. Kishore, B. Compton, N. Hmeidat, X. Chen, P. Liu, A. A. Hassen, J. Lindahl, V. Kunc, What makes a material printable? A viscoelastic model for extrusion-based 3D printing of polymers, J. Manuf. Process 35 (2018) 526–537, https://doi.org/10.1016/j.jmapro.2018.08.008.
[27] C. Ajinjeru, V. Kishore, J. Lindahl, Z. Sudbury, A.A. Hassen, B. Post, L. Love, V. Kunc, C. Duty, The influence of dynamic rheological properties on carbon fiber-reinforced polyetherimide for large-scale extrusion-based additive manufacturing, Int. J. Adv. Manuf. Technol. 99 (1) (2018) 411–418, https://doi.org/10.1007/s00170-018-2510-z.
[28] C.E. Duty, V. Kunc, B. Compton, B. Post, D. Erdman, R. Smith, R. Lind, P. Lloyd, L. Love, Structure and mechanical behavior of Big Area Additive Manufacturing (BAAM) materials, Rapid Prototyp. J. 23 (1) (2017) 181–189, https://doi.org/10.1108/RPJ-12-2015-0183.
[29] L.J. Love, V. Kunc, O. Rios, C.E. Duty, A.M. Elliott, B.K. Post, R.J. Smith, C.A. Blue, The importance of carbon fiber to polymer additive manufacturing, J. Mater. Res. 29 (17) (2014) 1893–1898, https://doi.org/10.1557/jmr.2014.212.
[30] ASTM D3171-15 Standard Test Methods for Constiuent Content of Composite Materials, ASTM International, West Conshohocken, PA, ASTM Standard, 2015, ⟨doi:10.1520/D3171–15⟩.
[31] Q. Wang, H. Ning, U. Vaidya, S. Pillay, L.-A. Nolen, Fiber content measurement for carbon fiber–reinforced thermoplastic composites using carbonization-in-nitrogen method, J. Thermoplast. Compos. Mater. 31 (1) (2018) 79–90, https://doi.org/10.1177/0892705716679481.
[32] G. Oliveux, L.O. Dandy, G.A. Leeke, Current status of recycling of fibre reinforced polymers: review of technologies, reuse and resulting properties, Prog. Mater. Sci. 72 (2015) 61–99, https://doi.org/10.1016/j.pmatsci.2015.01.004 (in English).